GRADE 4 SCIENCE

Chapter 1

Energy Causes Change

Teacher Packet Classroom Ready

Week of October 12-16, 2026

Big Question

Where can we observe evidence of energy causing change?

Teach	Project	Collect
Mon 10/12 and Tue 10/13; close Fri 10/16	Chapter 1 Slides (45 slides)	Guided Notes; Friday Check (16 points + percentage)

Chapter 2 begins Wednesday 10/14. Friday returns to Chapter 1 for recognition, reflection, and the scored check. Friday contains no first exposure to essential Chapter 1 science.

Classroom-ready wrapper of frozen Chapter 1 gold-standard version 1.1

Chapter Overview

Students learn that energy is the ability to cause change. They use observable before-and-after changes as evidence that energy was involved, identify five forms of energy, and explain that objects can have stored energy that can cause change later.

Field	Classroom Decision
Authoritative sources	CKSci Student Reader, Chapter 1, printed pages 1-6; Teacher Guide, Lessons 1-2
Core Reader pages	Pages 1-5; page 6 is a forward preview and is not required in Guided Notes
Big Question	Where can we observe evidence of energy causing change?
Core ideas	Energy can cause change; change is evidence; energy occurs in different forms; stored energy can cause change later; position can relate to stored energy
Vocabulary	energy; change; stored energy
Language to model	evidence; motion; sound; light; heat; electrical energy; position
Do not require	kinetic energy; potential energy; mechanical energy; radiant energy; joules; force; work equations

Learning Objectives

- Give an example of energy causing an observable change.
- Identify motion, sound, light, heat, and electrical energy in familiar examples.
- Explain that an object can have stored energy even when it is not moving.
- Use a before-and-after change as evidence that energy was involved.

Week Placement and Pacing

Day	Chapter	Required Sequence	Student Work	Time
Mon 10/12	Ch. 1 Session 1	Observe and Name: slides 1-3, 6-16	Notes p. 1: BQ, V1, N1	24.5 min
Tue 10/13	Ch. 1 Session 2	Forms and Stored Energy: slide 2, 17, 19-27	Notes p. 2: N2, V2, N3	24.5 min
Wed 10/14	Ch. 2 begins	Chapter 2 Session 1	Chapter 2 notes	24 min
Thu 10/15	Ch. 2	Chapter 2 Session 2 and quiz	Chapter 2 notes and quiz	24 min
Fri 10/16	Ch. 1 Session 3	Recognize, Reflect, Check: slide 2, 28-31, 33	Notes p. 3 and 16-point check	25 min

Required and Optional

Required	Recommended	Optional
Guided Notes; slides; electric pencil-sharpener demo; Friday check	Practice after Tuesday, before Friday	Energy Detectives after the check or during a later FLEX window; review slides 34-45

Friday rule: all vocabulary and core models are taught by Tuesday. Friday asks students to recognize and synthesize content already taught, then complete the scored check. Hand out the check no later than minute 12.

Session 1 Observe and Name

Monday October 12 | Goal: Students observe change, name it as evidence, and complete Guided Notes page 1.

Min	Slides	Teacher Move	Student Action
0-3	1-3	Introduce the Big Question. Ask for first thinking without correcting yet.	Turn and talk; write BQ first thinking.
3-8	6-9	Run the electric sharpener sequence: BEFORE, PREDICT, OBSERVE, AFTER, EVIDENCE.	Predict; watch; name visible and audible changes.
8-13	10-12	Teach V1 exactly: energy and change. Pause on the WRITE slide before reveal.	Fill change and different in V1.
13-17	13-14	Read and discuss Reader pages 1-2. Ask how a change becomes evidence.	Follow along; name one change.
17-21	15-16	Use N1 WRITE, then REVEAL. Complete ENERGY -> CHANGE -> EVIDENCE.	Fill evidence and complete the model.
21-24.5	-	Quick oral check and collect a few responses. Preserve the buffer.	Explain one observed change.

Ask and Listen For

Ask	Listen For
What changed?	The pencil became shorter and sharper; the sharpener changed from quiet to noisy.
What did you see or hear?	Specific observations, not a claim that the student saw energy.
What evidence tells us energy was involved?	The observable change is the evidence.

Optional slides 4-5 may be skipped freely. If slide 15 has not started by minute 24, move N1 to the first three minutes of Tuesday.

Session 2 Forms and Stored Energy

Tuesday October 13 | Goal: Students identify five forms of energy, explain stored energy, and complete Guided Notes page 2.

Min	Slides	Teacher Move	Student Action
0-1	2	Re-read the Big Question and ask what evidence the class has collected.	Recall Monday's evidence.
1-9	17, 19	Name the five forms exactly. Model one example, then release students to N2.	Write one change example for each form.
9-13.5	20-22	Teach V2 exactly. Pause on WRITE before reveal.	Fill later in the stored energy definition.
13.5-19.5	23-25	Show the still battery and a raised object. Use STORED ENERGY -> RELEASE -> CHANGE.	Predict a later change; discuss position.
19.5-24.5	26-27	Guide N3, then reveal samples. Correct only science, not personal example choice.	Fill position and record later changes.

Ask and Listen For

Ask	Listen For
Can an object have energy while it is still?	Yes. A battery, stretched rubber band, or raised object can have stored energy.
What could change later?	A flashlight can light; a rubber band can snap back; a raised object can fall.
Why does position matter?	A raised object can cause a change later when released.

Slide 18 is optional. If needed, complete only three N2 rows in class and finish the other two before Friday. All five rows are required by Friday.

Session 3 Recognize Reflect and Check

Friday October 16 | Goal: Students synthesize evidence already taught, answer the Big Question, and complete the 16-point Chapter 1 check.

Clock	Slides	Priority	Action
0:00-0:30	2	Required	Re-enter through the Big Question.
0:30-6:00	28-29	Priority 1	Complete N4 BEFORE/AFTER drawing and answer What changed?
6:00-9:30	30-31	Priority 3	Complete CHECK YOUR THINKING if time remains.
9:30-12:00	33	Priority 2	Write BQ NOW I THINK if it can finish before minute 12.
12:00-25:00	Check	Required	Administer all 9 items. Do not shorten the check.
25:00-30:00	-	Buffer	Collect, finish BQ/CHECK, or begin quiet optional work.

Friday Fallback

- N4 always happens in class.
- If N4 ends after minute 6, skip slides 30-31. CHECK YOUR THINKING moves to catch-up or home.
- If BQ NOW I THINK has not started by minute 9.5, complete it after the check or during catch-up.
- Hand out the check at minute 12 regardless of unfinished reflection work.

Optional slide 32 and review slides 34-45 are not part of the required Friday block.

Electric Pencil Sharpener Demonstration

Science focus: ENERGY -> CHANGE -> EVIDENCE. Do not explain the internal machinery or require advanced energy terms.

Stage	Teacher Action	Prompt	Expected Evidence
BEFORE	Show a dull pencil and a silent sharpener.	What do you notice before we begin?	Dull or rounded tip; pencil is longer; sharpener is quiet.
PREDICT	Hold the pencil near the sharpener without starting it.	What do you predict will change?	The tip may become sharper or shorter; the sharpener may make sound.
OBSERVE	Sharpen once while all students watch and listen.	What did you see or hear?	The pencil changed shape and length; the sharpener made noise.
AFTER	Hold the sharpened pencil beside the before pencil if available.	What is different now?	The tip is sharp; the pencil is slightly shorter; shavings may be visible.
EVIDENCE	Build the class model.	What evidence tells us energy was involved?	The observable changes are evidence that energy was involved.

Materials and Safety

- One electric pencil sharpener on a stable, visible surface.
- One dull pencil; an optional second already sharpened pencil for comparison.
- Teacher operates or directly supervises the sharpener. Keep fingers, hair, clothing, and loose items away from the opening.
- Do not claim heat unless the teacher actually observes it. Heat is not required for the demonstration.

If Students Ask

Where does the energy come from? Say: It comes through the cord or from batteries. We will study how energy moves and changes form in later chapters.

Guided Notes and Slide Anchor Map

When students write on Guided Notes, project the matching BQ, V, or N code. Slides without an anchor may support discussion but may not hide a required note answer.

Notes Anchor	Exact Student Task	Slides	Reveal
BQ	My first thinking; later, BQ NOW I THINK	2-3 and 33	Teacher discussion; no single answer
V1	energy - ability to cause change; change - become different	10-12	Slide 12: change, different
N1	When you see a change, you are seeing evidence of energy; complete model	13-16	Slide 16: evidence; ENERGY -> CHANGE -> EVIDENCE
N2	One example showing a change for each of five forms	17 and 19	Teacher checks examples
V2	stored energy can cause change at a later time	20-22	Slide 22: later
N3	Stored energy examples and position	23-27	Slide 27: position plus samples
N4	Draw BEFORE and AFTER; answer What changed?	28-31	Slide 31: sample response

Required Blanks in Order

V1: change, different | N1: evidence | V2: later | N3: position

HUD Labels

RECITE, WRITE, PREDICT, VOTE, TURN & TALK, CHECK, TRACE, DRAW, WATCH, OBSERVE, EVIDENCE, DATA, EXPLAIN are action labels. They remain separate from the semantic BQ/V/N anchor.

Teacher Questions Misconceptions and Responses

Student Idea	Teacher Response	What to Reinforce
I saw energy.	What did you actually see or hear change?	We observe the change. The change is evidence of energy.
Only moving objects have energy.	Can a battery cause a change later even while it is sitting still?	Objects can have stored energy.
Stored energy means it is moving.	Is the battery or raised book moving now? What could happen after release?	Stored energy can cause change later.
The desk is brown, so that is evidence.	Did the desk become brown during the observation?	A description is not automatically an observed change.
Proof and evidence are the same word here.	Accept proof once, then restate the class word evidence.	Use evidence in the Big Question and the N1 model.
The sharpener uses mechanical or kinetic energy.	Those terms are not needed in this chapter. What observable change did you notice?	Use the accepted Grade 4 terms.

High Value Prompts

- What changed in this specific object or system?
- What did it look or sound like before and after?
- Which form of energy is connected to the change?
- Why is that change evidence that energy was involved?
- Could this object cause a change later? What would release the stored energy?

What to Collect and How to Grade

Artifact	When	Status	Teacher Action
Guided Notes	Mon, Tue, Fri	Required; not scored	Check completion and science accuracy. Use the mirrored key.
Practice	After Tue; before Fri preferred	Recommended; ungraded	Review quickly or use the key for self-correction.
Chapter 1 Check	Fri minutes 12-25	Required; 16 points	Record raw score and whole-number percentage.
Energy Detectives	After the check or later FLEX	Optional; ungraded	Use look-fors only. Do not assign points.

Check Administration

- Choose OPEN NOTE or CLOSED NOTE before the check. If open note, allow only each student's own completed Guided Notes.
- Do not change the note policy during the assessment.
- Allow partial credit on items 3, 5, 6, 8, and 9 according to the key.
- Do not subtract for spelling or handwriting when the science idea is clear.

Raw Score to Percentage

Raw	%	Raw	%	Raw	%
16	100	10	63	4	25
15	94	9	56	3	19
14	88	8	50	2	13
13	81	7	44	1	6
12	75	6	38	0	0
11	69	5	31		

Key files: Guided_Notes_Key.pdf, Practice_Key.pdf, Check_Key.pdf. The optional Energy Detectives look-fors are in Energy_Detectives_Optional_Key.pdf.

Absent Student Procedure Extensions and Media

Absent Student

- Have the student read Student Reader pages 1-5 and complete Guided Notes in order: page 1, page 2, page 3.
- Use Reader examples and the written pencil-sharpener description. Do not ask the student to claim a live observation they did not make.
- Require the same check policy used with the class. Open note is allowed only if the class used open note.
- Optional Energy Detectives does not need to be made up.

Optional Energy Detectives

Use after the Friday check, on a review day, or during a later FLEX block. Students record an object or system, BEFORE, AFTER, the observable change, and the related form of energy. It is an ADAPTED extension from AP 1.1 and remains ungraded.

Optional Media Decision

The specifically named uploaded video was not present in the supplied attachment set. The closest repository micro-video was reviewed and is not recommended because it uses unsupported wording including zero state, null result, verified proof, and all physical actions. The lesson does not depend on video. Use the live pencil-sharpener demonstration.

Transition

Chapter 2 begins Wednesday October 14. Friday October 16 closes Chapter 1 through recognition, Big Question synthesis, and the scored check; it does not add new Chapter 1 science.

Classroom Ready Checklist and Provenance

Before Monday	Ready When
Print student packets	20 copies of the 7-page Student Print Packet
Print teacher materials	Teacher Packet, Materials and Prep, Guided Notes Key, Practice Key, Check Key
Project slides	Chapter_1_Slides.pptx opens; title slide is ready
Prepare demo	Sharpener works; dull pencil is ready; optional comparison pencil is ready
Choose check policy	OPEN NOTE or CLOSED NOTE is decided before Friday copies are used
Protect Friday timing	Check will be handed out no later than minute 12

Provenance

Tag	Use in this packet
SOURCE	Reader definitions, core science, Big Question, Teacher Guide demonstration and prompts
ADAPTED	Guided Notes models, practice, Chapter 1 check, Energy Detectives student page
CUSTOM	Classroom pacing wrapper, print directions, Friday fallback summary, score conversion table
EXTENSION	Energy Detectives as optional FLEX; everyday practice examples

Frozen Baseline

Gold-standard version 1.1 remains authoritative. This classroom-ready packet and bundle are derived wrappers. No frozen Chapter 1 file is modified in place.